Heavy-lift delivery drone

Saudi Arabia project | payload-release integration, remote field support and retrospective flight-log analysis

A distinct delivery platform, separate from the UAV-to-rover carrier and the DARPA Lift Challenge support work. The owner reports a 5 kg rice-payload delivery mission. The matching 21 January log independently establishes the travelled distance and control sequence.

20.25 km	10.11 km	35.67 min	5 kg
Total horizontal GPS travel	Maximum displacement from arming position	Armed duration, including takeoff and landing	Payload reported by the project owner



Supplied aircraft screenshot, rotated and cropped for presentation. The image identifies the delivery quadcopter; it is not a photograph of a weighed payload or a release event.

What the evidence establishes

The 21 January record contains a long outbound AUTO segment, a radio-failsafe transition to RTL, gripper command/state messages and a return close to the arming position. The approximately 20 km figure is the combined out-and-return distance, not a 20 km one-way radius.

The review also identifies vibration/clipping, a takeoff thrust-loss-check event and current-monitor data that cannot support an energy or endurance calculation. Findings are developed on the following pages.

Sources & review method

Separate early commissioning evidence from the later delivery-distance mission.

Evidence	7 January initial tests	21 January mission
Source filename	log_17_2026-1-7-10-40-56.bin	3. log_2_2026-1-21-21-36-00.bin
File size	83,543,263 bytes	61,440,893 bytes
Decoded messages	2,030,020	1,494,815
Record span	52.92 min	40.00 min
Armed periods	22.13 min and 17.51 min	35.67 min
GPS travel in armed periods	0.809 km and 0.640 km	20.249 km
Autopilot identity	Pixhawk6X / ArduCopter 4.6.3	Pixhawk6X / ArduCopter 4.6.3

Distance, timing and units

The DataFlash FMT records define each message layout. Parsing completed without discarded/resynchronization bytes. GPS instance 0, with a 3D-or-better fix, was selected within ARM-to-disarm windows. Distances are great-circle sums between consecutive fixes; no interval exceeded the 2-second/40 m/s rejection gates.

An independent trapezoidal integration of logged groundspeed gives 20.276 km, within 0.14% of the position-derived 20.249 km. Plots use elapsed time and distances only. Coordinates, satellite terrain, board identifiers and raw log attachments are excluded.

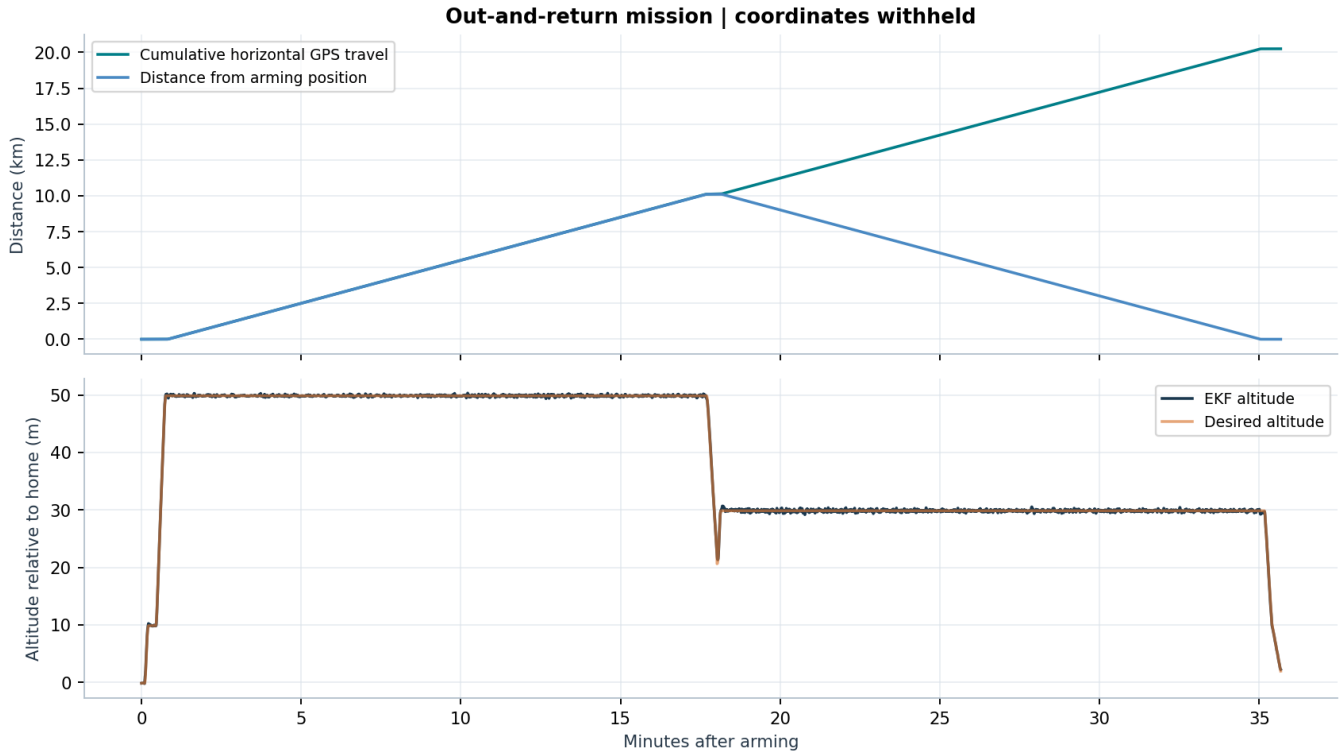
Hardware and claim boundaries

The log identifies a QUAD/X platform, Pixhawk6X and ArduCopter 4.6.3. The setup note describes connectivity, radio/sensor calibration, modes, failsafes, staged tests and log collection. SIYI MK32, Hobbywing X6 and charger manuals are reference documents; a manual alone does not prove a component was installed.

The 5 kg payload and Saudi Arabia project context come from Achraf RAHIBE. The log does not weigh the payload or independently observe the physical bag detaching. Command/state messages establish the gripper interface sequence.

Mission profile

Position-derived distance and EKF altitude, relative to the arming position and home datum.



Distance interpretation

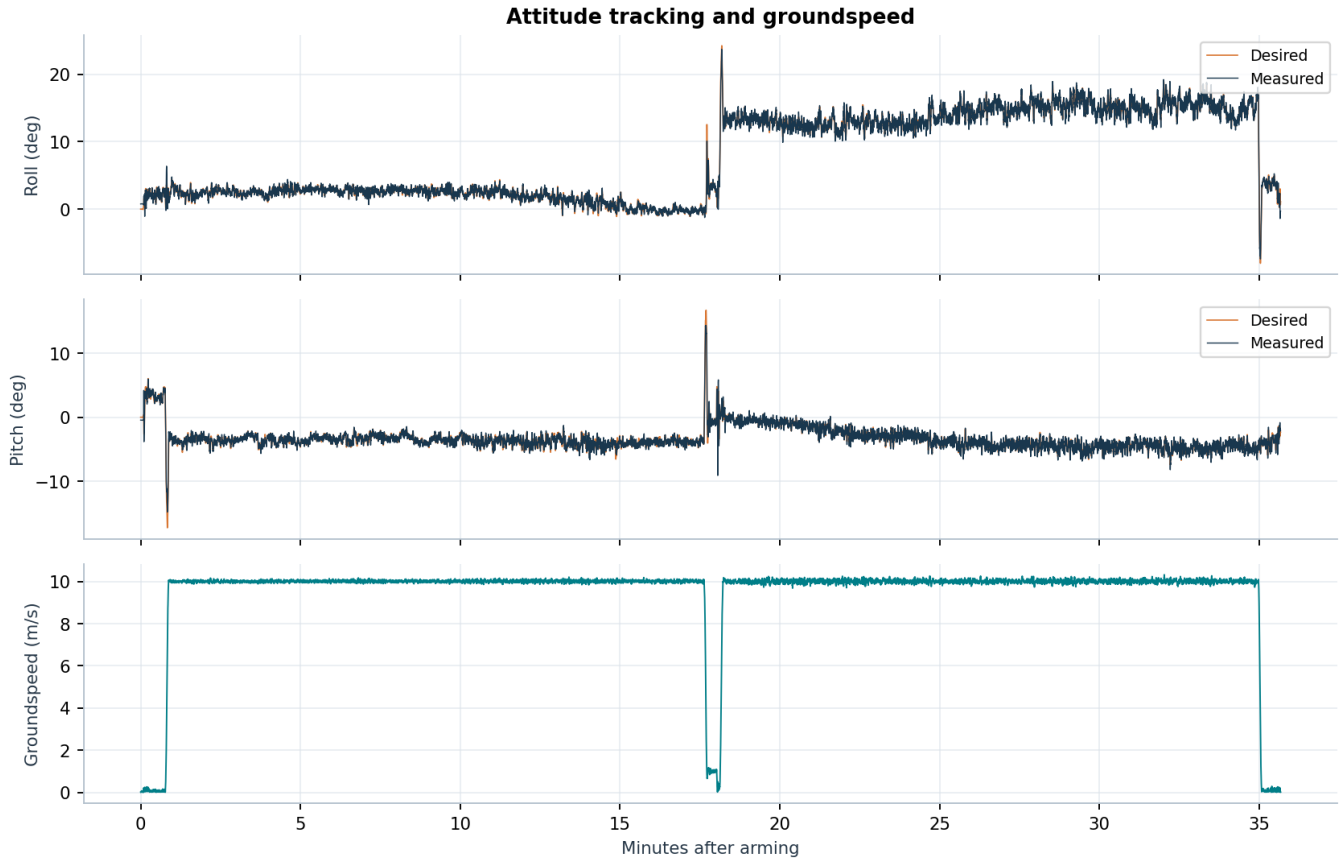
Total recorded GPS travel is 20.25 km. The furthest logged displacement from the arming position is 10.11 km; the return ends within approximately 0.1 m of that first fix. This last value reflects the logged positions and is not a claim of sub-metre navigation accuracy.

Maximum EKF altitude is 50.33 m relative to home. Median groundspeed during the armed interval is 10.01 m/s and maximum is 10.32 m/s. Relative-to-home altitude is not terrain-following AGL clearance.

Milestone	Minutes after arming
Arm / GUIDED takeoff	0.000
AUTO mission resumes	0.443
Radio failsafe / RTL	18.018
First logged release command	18.561
Low-battery event	35.549
Disarm	35.671

Flight-control tracking

Desired and measured attitude compared over the armed mission interval.



Metric	Roll	Pitch
Armed-window tracking RMSE	0.325 degrees	0.350 degrees

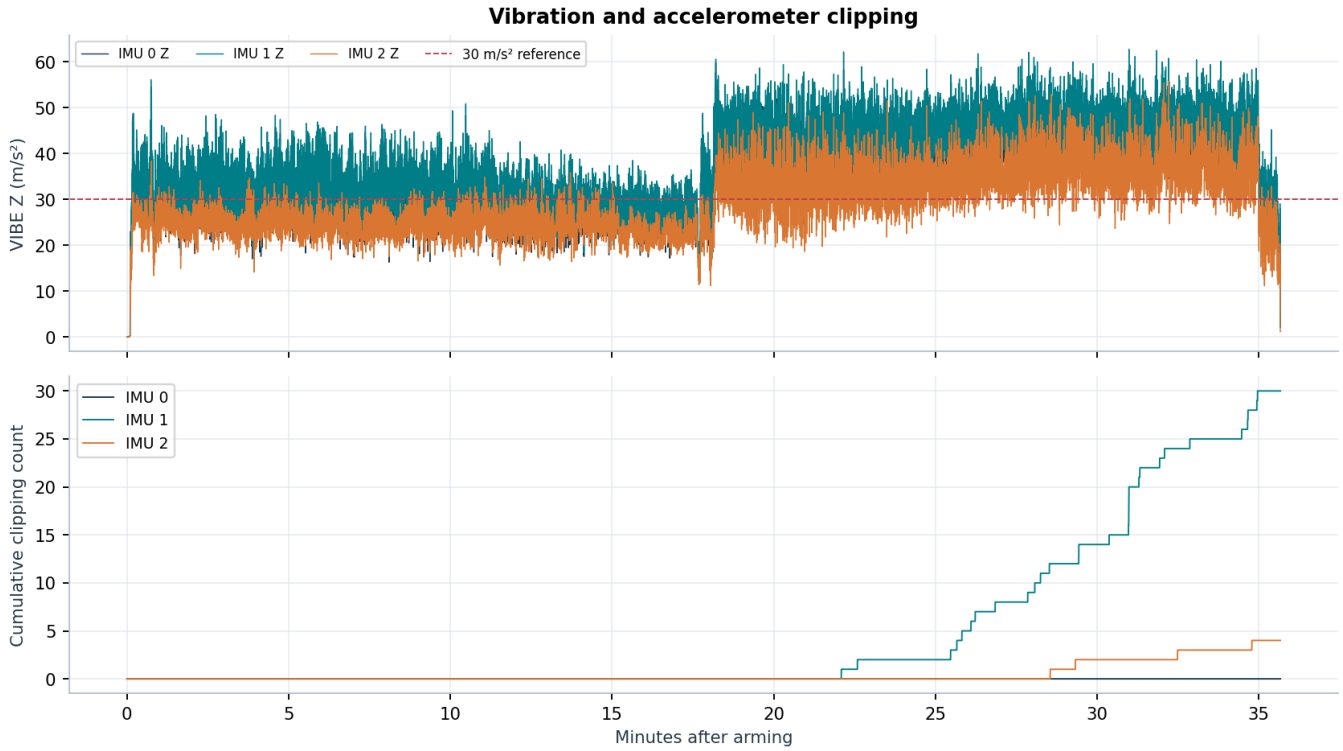
The desired and measured roll/pitch traces remain closely aligned over most of this record. Aggregate RMSE is descriptive of this flight, including takeoff and landing; it does not prove stability margins, optimal gains or all-condition performance.

An ERR 25/1 thrust-loss-check entry occurs approximately 5.6 seconds after arming, near takeoff. Review the short takeoff interval, propulsion response, loading and motor-output headroom before treating this event as resolved. The source does not establish its mechanical cause.

No revised PID values are prescribed from this retrospective alone. The next useful comparison is a controlled repeat with known payload, calibrated power telemetry and the same test gates.

Vibration & clipping

IMU-specific vibration and cumulative accelerometer-clipping evidence.



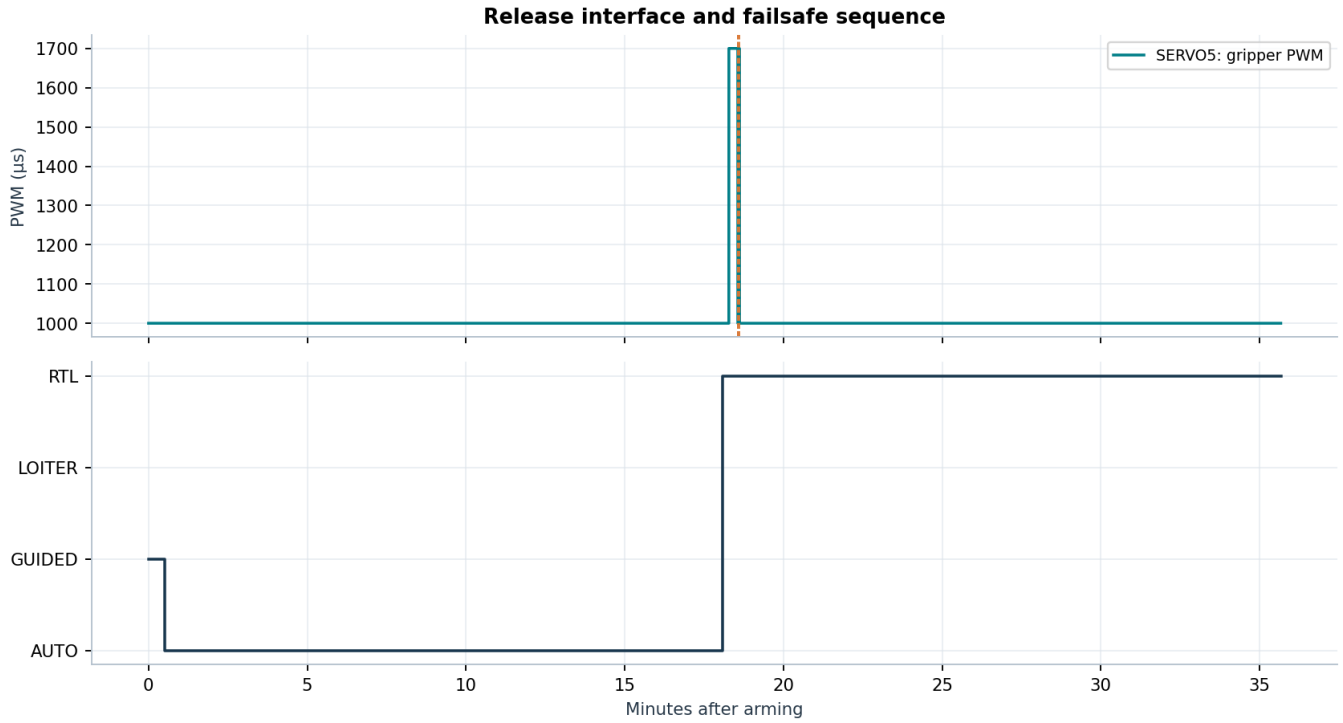
Mission interval	IMU 0	IMU 1	IMU 2
Z vibration, median (m/s²)	34.89	36.90	27.96
Z vibration, 95th percentile	48.42	51.05	41.09
Z vibration, maximum	59.81	62.79	55.85
Clipping count increase	0	30	4

IMU 0/1 Z vibration is above the ArduPilot 30 m/s² reference for a substantial part of the mission. IMU 1 briefly exceeds 60 m/s² and accumulates 30 clipping counts. The third IMU has lower median Z vibration but still records four clips.

These are follow-up findings, not evidence that a particular propeller, mount or filter caused the vibration. Compare mechanical balance, fasteners, sensor isolation and RPM/FFT evidence under a controlled payload. Confirm a reduced vibration baseline in another logged test.

Payload release & return logic

Servo output, gripper messages and flight-mode changes are kept in the same time frame.



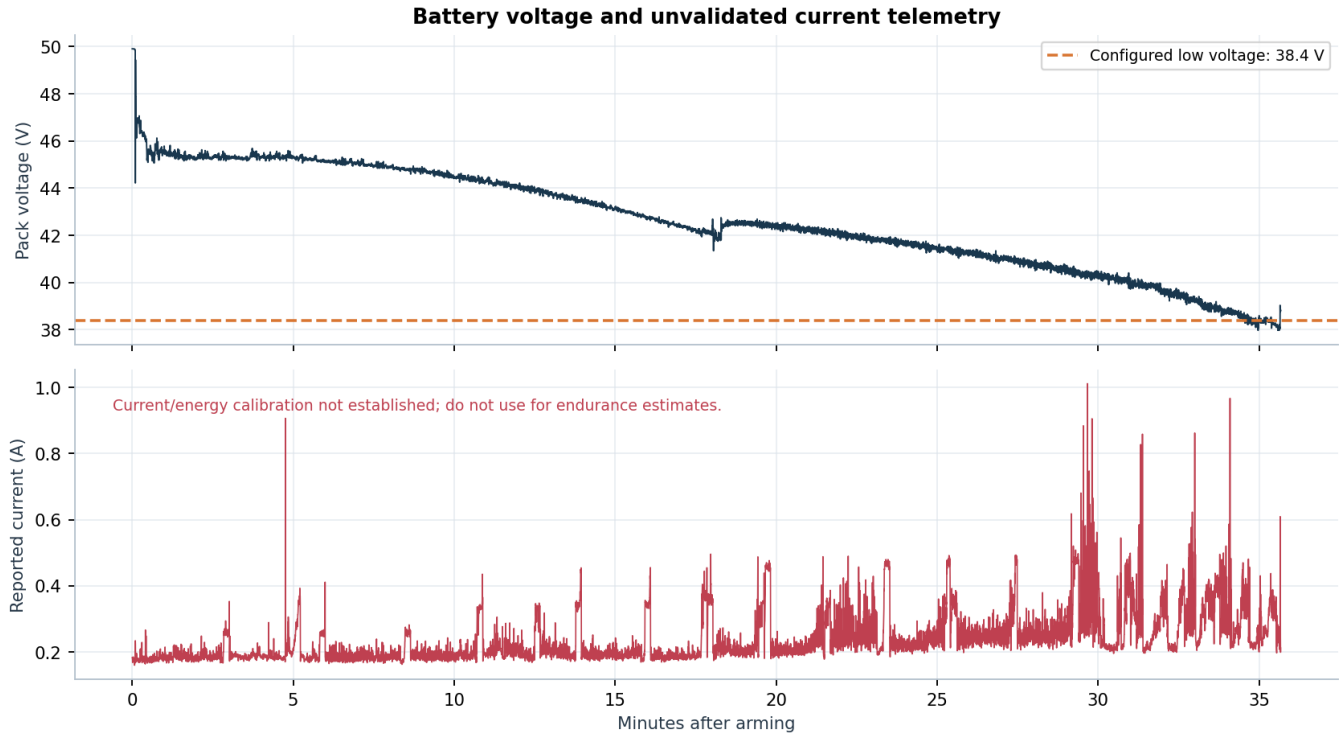
Configured interface	Recorded value
GRIP_ENABLE / GRIP_TYPE	1 / 1 (servo gripper)
SERVO5_FUNCTION	28 (gripper)
GRIP_GRAB / RELEASE / NEUTRAL	1700 / 1000 / 1500 microseconds
Release command times	18.561 and 18.613 min after arming

The log records grab/release commands and gripper state messages during the return sequence. The servo trace corroborates an actuated output path. It is not independent sensor confirmation that a 5 kg bag detached, reached a designated drop zone or remained attached for every kilometre.

Radio failsafe begins at 18.018 min and clears roughly 3.84 seconds later; the vehicle remains in RTL. Document the intended link-loss response and distinguish this recorded return from an uninterrupted, fully autonomous planned delivery-and-return sequence.

Power telemetry & test priorities

Voltage is reported separately from the current and consumed-energy channels.



Mission pack voltage ranges from 49.91 V to 37.97 V. A low-battery message reports 38.13 V at 35.549 min, shortly before disarm. The configured low-voltage threshold is 38.4 V. This record does not establish a calibrated remaining-energy reserve.

The logged current median is only 0.211 A and the maximum is 1.012 A. Consumed capacity rises by only about 141 mAh across a 35.7-minute heavy-lift flight. These readings are not physically credible as complete propulsion-current measurements for this aircraft. Calibration, scaling and sensor coverage must be verified; no Wh/km or payload-endurance figure is calculated.

Priorities for a repeatable validation run

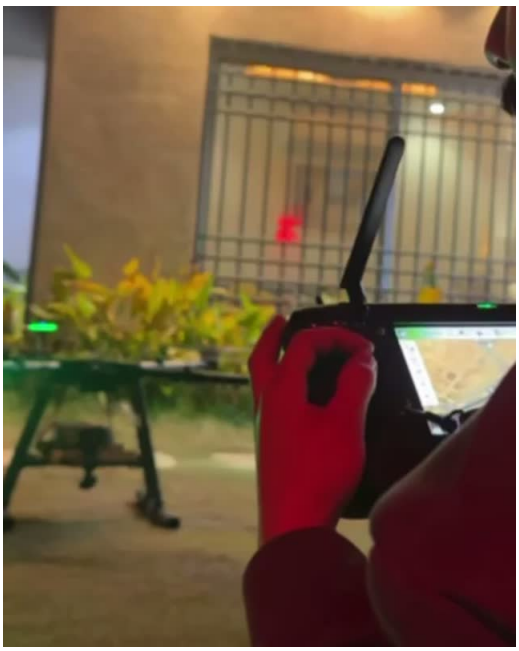
1. Verify current-monitor scaling and coverage against an independent reference; confirm the real battery capacity and reserve policy.

2. Investigate the takeoff thrust-loss-check event and measured vibration/clipping; retain before/after evidence rather than assuming a tuning fix.

3. Repeat with documented 5 kg mass, release-mechanism checks and visual release confirmation, while logging the intended radio-failsafe and return sequence.

Project screenshots & contribution

Remote field support, configuration and log review for this delivery platform.



Engineering contribution

Connectivity and ground-station setup; radio and sensor checks; autopilot configuration; staged field-test guidance; payload-release interfaces; flight-log review and next-test documentation.

The call excerpt shows remote support with the on-site operator. It is labeled as setup/test guidance, not as footage of the 20 km mission. Other aircraft in the supplied introduction montage are not assigned to this case.

Evidence retained for reproducibility

Early-test SHA-256: 7145e046401465cd817d5c27e4f355c9...
Mission SHA-256: 14af25b6deb4baa9045d0e769c33151b...
Full hashes, parser output and analysis scripts remain in the local review workspace. The public report excludes precise coordinates, raw binaries and private client identifiers.

References and interpretation

ArduPilot: onboard message definitions - field identities and units.

ArduPilot: diagnosing problems using logs - desired/measured tracking, vibration, clipping and power-review approach.

ArduCopter 4.6.3 gripper implementation - servo-gripper interface context.

Supplied project references: installation/setup note, SIYI MK32 manual v1.3, Hobbywing propulsion and battery-charger documents. Component references are not treated as measured aircraft performance.

Review scope: retrospective engineering evidence, not an airworthiness approval or a statement that the recorded follow-up findings were subsequently corrected.